

Explain the difference between the terms security, privacy and integrity of data

9618_w23_qp_12_sc_5.a

QUESTION

5 (a) State the meaning of **privacy of data**.

.....
..... [1]

MARK SCHEME

5(a)	1 mark for: Either - Ensuring data can only be accessed by / disclosed to authorised persons Or - Ensuring data cannot be accessed by / disclosed to unauthorised persons	1
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9618_w23_qp_12_sc_5.b

QUESTION

(b) State the meaning of **integrity of data**.

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..... [1]

MARK SCHEME

5(b)	1 mark for each bullet point (max 1) - Ensuring the accuracy / completeness / consistency of data (during / after processing) - Ensuring the data is up to date	1
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QUESTION

- (ii) State what is meant by data integrity **and** give **one** example of how this is implemented in a database.

Definition

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Example

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[2]

MARK SCHEME

2(a)(ii)	1 mark for definition - Methods of making sure the data is consistent 1 mark for example Examples: - Enforcing referential integrity - If data in one table is deleted/edited all tables are updated // cascading update/delete - Validation/verification rules	2
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QUESTION

- 1 When designing computer systems, it is important to consider the security, integrity and privacy of the data.

Draw **one** line from each measure to indicate whether it keeps data secure or protects the integrity of data.

Measure

[2]

MARK SCHEME

1	1 mark for 3 correct lines only from Data Security 1 mark for 2 correct lines only from Data Integrity The diagram shows the correct connections for the mark scheme. Firewall, Double entry, and Presence check are connected to Data Security. Access rights and Password are connected to Data Integrity.	2
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QUESTION

2 Xanthe wants to maintain the integrity and security of data stored on her computer.

(a) Explain the difference between data security and data integrity.

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..... [2]

MARK SCHEME

2(a)	1 mark per bullet point - security is protecting data from loss / corruption - integrity is ensuring the consistency / accuracy of the data	2
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QUESTION

8 A company has several security measures in place to prevent unauthorised access to the data on its computers.

(a) Describe the difference between the security and privacy of data.

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..... [2]

MARK SCHEME

8(a)	1 mark per bullet point - Security protects data against loss - Privacy protects data against unauthorised access	2
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Show appreciation of the need for both the security of data and the security of the computer system

Describe security measures designed to protect computer systems, ranging from the stand-alone PC to a network of computers

9618_w25_qp_12_sc_10.c

QUESTION

(c) A user downloads the computer program from the internet.

State what should be included as part of the download to make sure the program is authentic.

..... [1]

MARK SCHEME

10(c)	1 mark for a correct answer Digital Signature	1
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QUESTION

(b) The programmer needs to keep the program files secure on their computer and during electronic data transmission over the internet. The files are protected by a password.

Complete the table by identifying **one** other method of keeping the files secure during electronic data transmission **and one** other method of keeping the files secure on the computer.

State how each method protects the data.

The methods must be different.

	Method	How the method protects the data
during transmission		
		
		
on the computer		
		
		

[4]

MARK SCHEME

3(b)	1 mark for method and 1 mark for corresponding description During transmission e.g. • Encryption // by example such as VPN • Jumble / encode data so it cannot be decrypted/understood without the key On computer e.g. • Firewall / proxy • Filter incoming transmissions and stop any that could be attempting unauthorised access • Anti-malware • Find and delete or quarantine any malware that could delete the data / files • Encryption • Jumble / encode data so it cannot be decrypted / understood without the key • Physical method // by example • For example, the computer storing the data cannot be accessed without the key to the room	4
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9618_s25_qp_11_sc_7.b

QUESTION

(b) The banker receives confidential data across the internet. The data includes a digital signature.

Explain how a digital signature can make sure the data has **not** been changed during transmission.

[5]

MARK SCHEME

7(b)	1 mark each to max 5 • The sender hashes the document / message • to produce a digest • The sender encrypts the digest to create the digital signature • The message and the signature are sent to the banker / receiver • The receiver decrypts the signature to reproduce the digest • The receiver uses the same hashing algorithm on the document received to produce a second digest • The receiver compares this digest with the one from the digital signature • If both of the receiver's digests are the same the document has not changed	5
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9618_s24_qp_12_sc_3.a.i

QUESTION

- 3** An assessment board scans exam papers and stores the digitised papers on a server. Exam markers download the digitised papers to mark. The exam markers then upload the mark for each paper.

(a) The assessment board needs to make sure the data stored on the server is secure.

- (i)** Authentication methods can help to protect the server against hackers.

Identify **one other** security measure that helps to protect the server from hackers.

Describe how the security measure works.

Security measure

Description

.....

.....

.....

[3]

MARK SCHEME

3(a)(i)	1 mark for security measure 1 mark each to max 2 for how the chosen measure works: • Firewall • Checks incoming connections • ... against criteria • Blocks data from entering specific ports • Blocks data that does not meet whitelist that meets blacklist • Proxy server • Prevents devices accessing the web server directly • Intercepts any requests • Forwards the request using its own IP address • Screens returning data before sending it to the user	3
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9618_s24_qp_11_sc_5.c.i

QUESTION

(c) The bank also needs to keep its customers' data private and secure.

(i) The bank's network has a firewall.

Explain how a firewall can help protect the customers' data.

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..... [3]

MARK SCHEME

5(c)(i)	1 mark each to max 3: • Compares all incoming and outgoing transmissions • ... against set criteria/whitelist/blacklist • Blocks all transmissions that do not meet rules • Blocks data entering from specific ports • Blocks unauthorised/unknown internal software transmitting data	3
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QUESTION

- (ii) Customers need to use biometric authentication to access their accounts.
One biometric authentication method is facial recognition.

Facial recognition uses Artificial Intelligence (AI).

Describe how AI is used in facial recognition.

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..... [4]

MARK SCHEME

5(c)(ii)	1 mark each to max 4: e.g. • Captures an image of the face • Uses image recognition • Trained to identify the features of a face in an image • ... using a large number of images • Analyse images for facial features • Uses the probability of a match	4
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QUESTION

- (iii) State how the following security measures can be used to protect computer systems.

Firewall

.....

Encryption

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Passwords

.....

..... [3]

MARK SCHEME

3(a)(iii)	1 mark each for firewall, encryption and passwords. Firewall: • Monitors incoming and outgoing traffic and rejects any traffic that does not meet the set rules Encryption: • Ensures that if data is intercepted / obtained it cannot be understood without the decryption key Passwords: • Ensures only users with the correct password can access the resources // prevents unauthorised access	3
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9618_w23_qp_13_sc_8.c

QUESTION

(c) Explain how the data security risks of malware can be restricted.

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..... [3]

MARK SCHEME

8(c)	1 mark for each bullet point (max 3). • Download programs from reputable websites / sources • ...as these are less likely to contain malware • Backup / archive computer systems • ...so they can be restored in case of data loss from malware program installation • Install and run anti-malware program • ...so that regular scans can be made for known malware • ...and if malware is found it can be quarantined / removed • ...and computer's anti-malware definitions are regularly updated • Using a firewall to block unused ports • ...so that malware cannot enter the computer system • Deny administrator privileges to everyday users • ...so that malware cannot be downloaded by everyday users • Avoid the use of / access to removable devices • ...so that malware cannot be installed from these devices	3
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9618_s23_qp_13_sc_6.a

QUESTION

6 Data needs to be kept secure when stored on a computer and during transmission over a network.

(a) Explain how a digital signature is used to authenticate a digital document during transmission over a network.

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..... [5]

MARK SCHEME

6(a)	1 mark each to max 5 • The sender hashes the document • ... to produce a <u>digest</u> • The sender <u>encrypts</u> the digest to create the digital signature • The message and the signature are sent to the receiver • The receiver <u>decrypts</u> the signature to reproduce the digest • The receiver uses the <u>same</u> hashing algorithm on the document received to produce a second digest • The receiver compares this digest with the one from the digital signature • If both of the receiver's digests are the same the document is authentic	5
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9618_s23_qp_13_sc_6.b

QUESTION

- (b) Complete the table by identifying **and** describing **two** types of software that can be installed on a computer to prevent threats over a network.

Type of software	Description
.....	
.....	

[2]

MARK SCHEME

6(b)	1 mark each for identification and appropriate description of 2 pieces of software, max 2 <table><tr><th>Type of software</th><th>Description</th></tr><tr><td>Antivirus</td><td> - scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing </td></tr><tr><td>Antispyware</td><td> - scans the computer for spyware and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known spyware and prevents the download continuing. </td></tr><tr><td>Firewall</td><td> - monitors incoming and outgoing traffic and compares it to criteria that are set by the user such as through a whitelist/blacklist/identifying allowed / blocked IP addresses - compares incoming and outgoing traffic to criteria blocks those that do not match criteria </td></tr><tr><td>Antimalware</td><td> - scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing </td></tr></table>	Type of software	Description	Antivirus	- scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing	Antispyware	- scans the computer for spyware and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known spyware and prevents the download continuing.	Firewall	- monitors incoming and outgoing traffic and compares it to criteria that are set by the user such as through a whitelist/blacklist/identifying allowed / blocked IP addresses - compares incoming and outgoing traffic to criteria blocks those that do not match criteria	Antimalware	- scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing	2
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9618_w22_qp_12_sc_6.a.i

QUESTION

6 (a) A student uses a networked laptop computer to send an email to a colleague.

(i) Explain how a digital signature ensures the email is authentic.

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..... [2]

MARK SCHEME

6(a)(i)	1 mark for each method of ensuring authenticity (max 2): • (email) message put through hashing algorithm to produce a digest • Digest encrypted with <u>sender's private</u> key (to create the digital signature) • the (digital) signature can only be decrypted with matching <u>sender's public</u> key	2
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9618_w22_qp_12_sc_6.a.ii

QUESTION

(ii) Describe how a firewall protects the data on the computer.

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..... [3]

MARK SCHEME

6(a)(ii)	1 mark for each bullet point: • monitors <u>incoming</u> and <u>outgoing</u> packets / traffic • checks against an allow list / deny list of IP addresses // checks against a set of rules for acceptable data / ports etc. • blocks transmissions that do not meet criteria / rules // allows through is satisfies the criteria /rules	3
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QUESTION

- 2 Draw **one** line from each security feature to its most appropriate description.

Security feature	Description
firewall	converts data to an alternative form
pharming	redirects a user to a fake website
anti-virus software	verifies the authenticity of data
encryption	scans files on the hard drive for malicious software
	accepts or rejects incoming and outgoing packets based on criteria

[4]

MARK SCHEME

2	1 mark for each correct line. Security feature firewall pharming anti-virus software encryption converts data to an alternative form redirects a user to a false website verifies the authenticity of data scans files on the hard drive for malicious software accepts or rejects incoming and outgoing packets based on criteria	4
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9618_s21_qp_12_sc_8.b

QUESTION

- (b) Each employee has a username and password to allow them to log onto a computer. An employee's access rights to the data on the computers is set to either read-only, or read and write.

Identify **one** other software-based measure that could be used to restrict the access to the data on the computers.

.....
 [1]

MARK SCHEME

8(b)	1 mark for a correct answer • Two factor authentication • Biometric passwords • Key Card Access • Firewall	1
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Show understanding of the threats to computer and data security posed by networks and the internet

9618_w25_qp_12_sc_5.e

QUESTION

- (e) Complete the table by identifying **one** threat to computer and data security posed by networks and the internet.

Describe the threat and give a method of prevention.

Threat	Description	Prevention
		
		
.....		
.....		
		
		

[3]

MARK SCHEME

5(e)	1 mark for correct threat 1 mark corresponding description 1 mark for correct method of prevention <table border="1"> <thead> <tr> <th>Threat</th><th>Description</th><th>Prevention</th></tr> </thead> <tbody> <tr> <td>Malware / Virus</td><td>Malicious code that can alter/delete files</td><td>Anti-virus // Anti-malware // Firewall</td></tr> <tr> <td>Spyware</td><td>Records keystrokes which are sent to third party</td><td>Anti-spyware // Firewall // Anti-malware</td></tr> <tr> <td>Hacking</td><td>Gaining <u>unauthorised</u> access to a computer network / device</td><td>Authentication // Firewall</td></tr> <tr> <td>Phishing</td><td><u>Emails</u> supposedly from reputable companies are sent to trick people into revealing personal information</td><td>Use a spam filter // Do not open emails from unknown sources</td></tr> <tr> <td>Pharming</td><td>Users are directed to a bogus <u>website</u> that looks legitimate to obtain personal information</td><td>Use a VPN // Anti-malware// Do not open links or download attachments</td></tr> </tbody> </table>	Threat	Description	Prevention	Malware / Virus	Malicious code that can alter/delete files	Anti-virus // Anti-malware // Firewall	Spyware	Records keystrokes which are sent to third party	Anti-spyware // Firewall // Anti-malware	Hacking	Gaining <u>unauthorised</u> access to a computer network / device	Authentication // Firewall	Phishing	<u>Emails</u> supposedly from reputable companies are sent to trick people into revealing personal information	Use a spam filter // Do not open emails from unknown sources	Pharming	Users are directed to a bogus <u>website</u> that looks legitimate to obtain personal information	Use a VPN // Anti-malware// Do not open links or download attachments	3
Threat	Description	Prevention																		
Malware / Virus	Malicious code that can alter/delete files	Anti-virus // Anti-malware // Firewall																		
Spyware	Records keystrokes which are sent to third party	Anti-spyware // Firewall // Anti-malware																		
Hacking	Gaining <u>unauthorised</u> access to a computer network / device	Authentication // Firewall																		
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Pharming	Users are directed to a bogus <u>website</u> that looks legitimate to obtain personal information	Use a VPN // Anti-malware// Do not open links or download attachments																		

9618_w25_qp_11_sc_8.a

QUESTION

- 8** One ethical consideration for a student connecting their personal computer to the school network is the risk of spreading malware on the network.

- (a)** Viruses and pharming are examples of malware.

Explain what is meant by a virus and pharming.

Virus

.....

.....

Pharming

.....

.....

[2]

MARK SCHEME

8(a)	1 mark for each correct explanation, max 2 marks Virus: Malicious program/software that replicates/copies itself and deletes/alters files/data stored on a computer Pharming: Malicious code/software installed on a computer which redirects user to a fake website to obtain personal data	2
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9618_w23_qp_13_sc_8.b

QUESTION

(b) State **one** difference and **one** similarity between pharming and phishing.

Difference

.....

Similarity

.....

[2]

MARK SCHEME

8(b)	1 mark for difference 1 mark for similarity Difference: - Pharming is malicious code that redirects to a fake website. Phishing uses an email to prompt user action. - Pharming is automatic. Phishing requires user action. Similarity: - Both try to obtain financial or personal information - Both are a false representation of an official organisation, e.g. a bank - Both make use of fake websites	2
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QUESTION

(c) Describe the following threats to a computer system.

Phishing email

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.....

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Spyware

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.....

[4]

MARK SCHEME

5(c)	1 mark for each bullet point. Phishing email (max 2) • The email pretends to be from an official body • ... persuading individuals to disclose private information // by example such as bank details • ... or requesting authentication by redirecting to an unofficial/unauthorised website // inviting a user to click a link Spyware (max 2) • Malware downloaded without the user's knowledge • ... which secretly records the user's actions / keystrokes on the computer • ... and sends logs of the actions to a third party	4
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QUESTION

- 2 Draw **one** line from each security feature to its most appropriate description.

Security feature	Description
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MARK SCHEME

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9618_w21_qp_11_sc_2.c

QUESTION

(c) Two malware threats are spyware and viruses.

Give **two** similarities and **one** difference between spyware and a virus.

Similarity 1

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Similarity 2

.....

Difference

.....

[3]

MARK SCHEME

2(c)	1 mark per similarity to max 2 - Both are pieces of malicious software - Both are downloaded / installed/run without the user's knowledge - Both can pretend to be / are embedded in other legitimate software when downloaded // both try to avoid the firewall - Both run in the background 1 mark for difference - Virus can damage computer data; spyware only records / accesses data - Virus does not send data out of the computer; spyware sends recorded data to third party - Virus replicates itself; spyware does not replicate itself	3
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9618_s21_qp_12_sc_8.c

QUESTION

(c) The company is also concerned about threats posed by networks and the internet.

Identify **two** threats to the data that are posed by networks and the internet.

Threat 1

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Threat 2

.....

[2]

MARK SCHEME

8(c)	1 mark per correct answer to max 2 - Malware // viruses // spyware // by example - Hacking - Phishing - Pharming	2
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Describe methods that can be used to restrict the risks posed by threats

9618_w25_qp_12_sc_5.e

QUESTION

- (e) Complete the table by identifying **one** threat to computer and data security posed by networks and the internet.

Describe the threat and give a method of prevention.

Threat	Description	Prevention
		
		
.....		
.....		
		
		

[3]

MARK SCHEME

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9618_s25_qp_11_sc_7.a

QUESTION

7 A banker stores personal data on their work computer.

(a) The banker needs to transfer confidential data across the internet.

Identify and describe **one** method of restricting the risks posed by an unauthorised person intercepting the data whilst it is being transferred across the internet.

Method

Description

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.....

.....

.....

[3]

MARK SCHEME

7(a)	1 mark for a correct method: 1 mark each to max 2 for a corresponding description Method: Encryption Description: • Data is encoded/scrambled using a key to create cipher text • if intercepted it cannot be understood • without being decrypted using a key	3
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9618_s24_qp_12_sc_3.a.i

QUESTION

3 An assessment board scans exam papers and stores the digitised papers on a server. Exam markers download the digitised papers to mark. The exam markers then upload the mark for each paper.

(a) The assessment board needs to make sure the data stored on the server is secure.

(i) Authentication methods can help to protect the server against hackers.

Identify **one other** security measure that helps to protect the server from hackers.

Describe how the security measure works.

Security measure

Description

.....

.....

.....

[3]

MARK SCHEME

3(a)(i)	1 mark for security measure 1 mark each to max 2 for how the chosen measure works: • Firewall • Checks incoming connections • ... against criteria • Blocks data from entering specific ports • Blocks data that does not meet whitelist that meets blacklist • Proxy server • Prevents devices accessing the web server directly • Intercepts any requests • Forwards the request using its own IP address • Screens returning data before sending it to the user	3
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QUESTION

(c) Explain how the data security risks of malware can be restricted.

.....

.....

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..... [3]

MARK SCHEME

8(c)	1 mark for each bullet point (max 3). • Download programs from reputable websites / sources • ...as these are less likely to contain malware • Backup / archive computer systems • ...so they can be restored in case of data loss from malware program installation • Install and run anti-malware program • ...so that regular scans can be made for known malware • ...and if malware is found it can be quarantined / removed • ...and computer's anti-malware definitions are regularly updated • Using a firewall to block unused ports • ...so that malware cannot enter the computer system • Deny administrator privileges to everyday users • ...so that malware cannot be downloaded by everyday users • Avoid the use of / access to removable devices • ...so that malware cannot be installed from these devices	3
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QUESTION

- (b) Complete the table by identifying **and** describing **two** types of software that can be installed on a computer to prevent threats over a network.

Type of software	Description
.....	
.....	

[2]

MARK SCHEME

6(b)	1 mark each for identification and appropriate description of 2 pieces of software, max 2 <table><tr><th>Type of software</th><th>Description</th></tr><tr><td>Antivirus</td><td> - scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing </td></tr><tr><td>Antispyware</td><td> - scans the computer for spyware and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known spyware and prevents the download continuing. </td></tr><tr><td>Firewall</td><td> - monitors incoming and outgoing traffic and compares it to criteria that are set by the user such as through a whitelist/blacklist/identifying allowed / blocked IP addresses - compares incoming and outgoing traffic to criteria blocks those that do not match criteria </td></tr><tr><td>Antimalware</td><td> - scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing </td></tr></table>	Type of software	Description	Antivirus	- scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing	Antispyware	- scans the computer for spyware and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known spyware and prevents the download continuing.	Firewall	- monitors incoming and outgoing traffic and compares it to criteria that are set by the user such as through a whitelist/blacklist/identifying allowed / blocked IP addresses - compares incoming and outgoing traffic to criteria blocks those that do not match criteria	Antimalware	- scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing	2
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Describe security methods designed to protect the security of data

9618_s25_qp_13_sc_3.b

QUESTION

(b) The programmer needs to keep the program files secure on their computer and during electronic data transmission over the internet. The files are protected by a password.

Complete the table by identifying **one** other method of keeping the files secure during electronic data transmission **and one** other method of keeping the files secure on the computer.

State how each method protects the data.

The methods must be different.

	Method	How the method protects the data
during transmission		
		
		
on the computer		
		
		

[4]

MARK SCHEME

3(b)	1 mark for method and 1 mark for corresponding description During transmission e.g. • Encryption // by example such as VPN • Jumble / encode data so it cannot be decrypted/understood without the key On computer e.g. • Firewall / proxy • Filter incoming transmissions and stop any that could be attempting unauthorised access • Anti-malware • Find and delete or quarantine any malware that could delete the data / files • Encryption • Jumble / encode data so it cannot be decrypted / understood without the key • Physical method // by example • For example, the computer storing the data cannot be accessed without the key to the room	4
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9618_s25_qp_12_sc_5.c.i

QUESTION

(c) A Database Management System (DBMS) provides data security.

- (i) Identify **two** methods the DBMS can use to protect the data in the table `USER` from unauthorised access.

Explain how each method protects the data.

Method 1

Explanation

.....

.....

Method 2

Explanation

.....

.....

[4]

MARK SCHEME

5(c)(i)	1 mark for method 1 mark for matching explanation e.g. • Access rights • appropriate permissions for the table USER are needed to read or edit the data • A password for the database or the USER table • prevents users without the password from accessing the data • Encrypting the database • stops users without the decryption key from decoding / understanding the data • Views • users can be given a view to the database that does not include the data in the table USER	4
---------	---	----------

9618_s25_qp_11_sc_7.a

QUESTION

7 A banker stores personal data on their work computer.

(a) The banker needs to transfer confidential data across the internet.

Identify and describe **one** method of restricting the risks posed by an unauthorised person intercepting the data whilst it is being transferred across the internet.

Method

Description

.....

.....

.....

.....

[3]

MARK SCHEME

7(a)	1 mark for a correct method: 1 mark each to max 2 for a corresponding description Method: Encryption Description: • Data is encoded/scrambled using a key to create cipher text • if intercepted it cannot be understood • without being decrypted using a key	3
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QUESTION

(c) The file containing the final program code will be sent by email for beta testing.

Identify **one** security method that can be used to protect the program code from unauthorised access during email transfer.

Explain how your chosen method protects the program code.

Security method

.....

Explanation

.....

.....

.....

[3]

MARK SCHEME

4(c)	1 mark for the security method. 2 marks for explanation Security method: Encryption Explanation • File contents are converted to cipher text • If intercepted the data cannot be understood without the decryption key	3
------	---	----------

QUESTION

(ii) Explain how encryption can protect the security of data during transmission.

.....

.....

.....

..... [2]

MARK SCHEME

7(f)(ii)	1 mark each to max 2 : • Encodes/scrambles data • ... so if it is intercepted it cannot be understood • Algorithm/key is required to decode the data	2
----------	--	----------

9618_s24_qp_12_sc_3.a.ii

QUESTION

- (ii) Identify **one** security measure that helps to protect the data when it is being transmitted to its destination. Describe how the security measure works.

Security measure

Description

.....

[3]

MARK SCHEME

3(a)(ii)	1 mark for security measure 1 mark each to max 2 for description of the chosen measure: • Encryption • Encodes/scrambles data • ... so if it is intercepted it cannot be understood • Algorithm/key is required to decode the data	3
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9618_w21_qp_11_sc_5.b

QUESTION

- (b) Describe the ways in which access rights can be used to protect the data in Javier's database from unauthorised access.

.....

 [3]

MARK SCHEME

5(b)	1 mark per bullet point • Access rights give managers / himself access to different elements • ... by having different accounts / logins • ... which have different access rights e.g. read only // no access / read / write • Specific <u>views</u> can be assigned to himself and to the managers • ... e.g. managers can only see the data for their own shop(s)	3
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9618_s21_qp_12_sc_8.b

QUESTION

- (b) Each employee has a username and password to allow them to log onto a computer. An employee's access rights to the data on the computers is set to either read-only, or read and write.

Identify **one** other software-based measure that could be used to restrict the access to the data on the computers.

.....

..... [1]

MARK SCHEME

8(b)	1 mark for a correct answer • Two factor authentication • Biometric passwords • Key Card Access • Firewall	1
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